

Naman Gupta

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SUMMARY

Second-year B.Tech student at BIT Mesra with practical experience building and deploying machine learning systems on real-world structured and text datasets. Developed production-grade pipelines on datasets up to 280K samples, achieving 0.96 ROC AUC and 92% classification accuracy using PyTorch, Hugging Face, and FastAPI. Eager to tackle high-impact data problems and deliver solutions that go beyond the notebook.

PROJECTS

SentinelAI | *Python, PyTorch, Hugging Face Transformers, FastAPI, scikit-learn*

[🔗 Code](#)

- Fine-tuned **BERT-based transformer models** on **60K+ labeled social media comments** for hate speech detection using PyTorch and Hugging Face Transformers.
- Cleaned and tokenized 60K+ raw social media comments, standardizing noisy text input before feeding it into the transformer pipeline.
- Achieved **92% classification accuracy** using contextual embeddings from transformer models, outperforming baseline TF-IDF approaches.
- Deployed a **FastAPI inference API** for real-time toxicity prediction, enabling low-latency scoring on incoming text requests.

FraudShield | *Python, scikit-learn, CatBoost, SMOTE, FastAPI*

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- Worked with a highly imbalanced **280K+ transaction dataset** to flag fraudulent activity, handling skewed class distribution as the core technical challenge.
- Engineered **8+ behavioral transaction features** and applied SMOTE resampling, significantly improving minority-class recall for fraud detection.
- Trained and evaluated **CatBoost and gradient boosting models**, achieving **0.96 ROC AUC** on held-out validation data.
- Deployed the model using **FastAPI** and integrated a **Streamlit frontend**, enabling real-time and batch fraud predictions via a live web interface.

CredShield | *Python, scikit-learn, CatBoost, Flask*

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- Built a credit risk prediction model on **30K+ borrower records**, leveraging repayment history and credit utilization to capture key default risk signals.
- Engineered **financial risk features** including repayment behavior, credit utilization, and debt ratios to enhance model signal quality.
- Trained and evaluated **Logistic Regression and CatBoost models**, achieving **0.91 ROC AUC** through comparative performance analysis.
- Optimized classification threshold using **precision-recall analysis** to maximize **F1-score**, balancing false positives against missed high-risk defaults.

TECHNICAL SKILLS

Languages: Python, C++

Frameworks & Libraries: PyTorch, Hugging Face Transformers, scikit-learn, CatBoost, NumPy, Pandas, Matplotlib, Seaborn

Skills: Supervised Learning, Feature Engineering, EDA, Transformers, Transfer Learning, CNNs, RNNs, Text Classification

Databases: SQL, PostgreSQL

Deployment: FastAPI, AWS, Google Cloud, Render

Tools: Git, GitHub, Jupyter Notebook, Tableau

EDUCATION

Birla Institute of Technology, Ranchi

Aug 2024 – Aug 2028

B.Tech in Electronics and Communication

CGPA: 8.2/10

POSITIONS OF RESPONSIBILITY

Entrepreneurship Development Cell, BIT Mesra

2025 – Present

Core Team Member | E-Summit'26

- Led outreach, sponsorship, and logistics for **E-Summit'26**, an entrepreneurship event with **8000+** participants.
- Coordinated with startup founders, sponsors, and internal teams for speaker onboarding, partnerships, and event execution.